

Bayesian Regularized Kernel Machine

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1 The Model

The model under consideration is a kernel machine regularized through Bayesian priors on model parameters:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \dots + \beta_L y_{t-L} + x_t' \gamma + f(z_t) + \epsilon_t \quad (1)$$

for $t = L+1, \dots, T$, where the nonlinear component $f(\cdot)$ follows a Gaussian process with the Gaussian kernel matrix K whose (i, j) element is $\sigma^2 \exp(-\psi(z_i - z_j)'(z_i - z_j))$ for $i, j = 1, 2, \dots$ and the residual ϵ_t follows an i.i.d. conditional Gaussian distribution $\epsilon_t | s \sim N(0, s)$.

For sampling efficiency, re-write $f(z_t) = \sigma g(z_t)$ where $g(\cdot)$ follows a Gaussian process with the re-scaled Gaussian kernel matrix K_g whose (i, j) element is $\exp(-\psi(z_i - z_j)'(z_i - z_j))$. Let $n = T - L$, $\alpha = [\beta_0 \ \beta_1 \ \dots \ \beta_L \ \gamma']'$, $\beta = [\beta_1 \ \dots \ \beta_L \ \gamma' \ \sigma']'$, $g = [g(z_{L+1}) \ \dots \ g(z_T)]'$,

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$\tilde{\psi} = \pm\sqrt{\psi}$ and m be the length of the vector β . The priors are:

$$\begin{aligned}
s|s_0 &\sim IG(1/2, 1/s_0), \quad s_0 \sim IG(1/2, 1), \\
\beta_0 &\sim N(0, v_{\alpha,0}), \\
\beta|\tau, \lambda &\sim N(0, \text{diag}(\tau\lambda)), \quad \lambda = [\lambda_1 \dots \lambda_m]' \\
\tau|\tau_0 &\sim IG(1/2, 1/\tau_0), \quad \tau_0 \sim IG(1/2, n^2), \\
\lambda_j|\lambda_{j,0} &\sim IG(1/2, 1/\lambda_{j,0}), \quad \lambda_{j,0} \sim IG(1/2, 1), \quad j = 1, \dots, m, \\
\tilde{\psi} &\sim N(0, \lambda_\psi), \quad \lambda_\psi \sim IG(1/2, 1/\lambda_{\psi,0}), \quad \lambda_{\psi,0} \sim IG(1/2, 1), \\
g &\sim N(0, K_g)
\end{aligned} \tag{2}$$

where $N(\cdot, \cdot)$ denotes the Gaussian distribution, $G(\cdot, \cdot)$ denotes the gamma distribution of the shape-scale form, and $IG(\cdot, \cdot)$ denotes the inverse gamma distribution. Note that the horseshoe prior of Carvalho et al. (2010) is applied for β and $\pm\sqrt{\psi}$ to enforce shrinkage on the insignificant members of these parameters. Also note that the global shrinkage parameter τ is further regularized following Piironen and Vehtari (2017). The intercept β_0 has no shrinkage with a constant prior variance $v_{\alpha,0}$ (e.g. 10). As for the residual variance s , a gamma prior is placed for its inverse $1/s \sim G(1/2, 2\mu_s)$ such that $E(1/s) = \mu_s$. Given the uncertainty about the appropriate value of μ_s , an inverse gamma prior $\mu_s \sim IG(1/2, 1/2)$ is placed. Setting $s_0 = 2\mu_s$ gives the prior for s shown in Equation (2).

2 Estimation

Let $\theta = [s \ s_0 \ \alpha' \ \sigma \ \tilde{\psi} \ \tau \ \tau_0 \ \lambda' \ \lambda_{1,0} \ \dots \ \lambda_{m,0} \ \lambda_\psi \ \lambda_{\psi,0}]'$ collect model parameters, $f = [f(z_{L+1}) \ \dots \ f(z_T)]'$ collect the latent nonlinear components and $y = [y_1 \ \dots \ y_T]'$ be the data sample. The posterior distribution of interest is $p(\theta, f|y)$ and is sampled by a Gibbs sampler¹. The estimation steps are as follows:

1. Draw $\theta^{(0)}$ and $f^{(0)}$ from their priors.

¹For expositional convenience, the regressors x_t and z_t are omitted from the conditioning information set in the posterior distribution.

2. Design a for loop with M sweeps.

3. Within sweep $j \in \{1, 2, \dots, M\}$, do the following:

(a) Draw $s^{(j)}$ from $IG(1/2 + n/2, 1/s_0 + \epsilon'\epsilon/2)$, where $s_0 = s_0^{(j-1)}$, $\epsilon = [\epsilon_{L+1} \dots \epsilon_T]'$ is computed based on draws $\theta^{(j-1)}$ and $f^{(j-1)}$ in the previous sweep.

(b) Draw $s_0^{(j)}$ from $IG(1, 1 + 1/s)$ where $s = s^{(j)}$.

(c) Draw $\alpha^{(j)}$ from $N(b_\alpha, B_\alpha)$, where $B_\alpha^{-1} = \Sigma_0^{-1} + \tilde{X}'(sI_n + \sigma^2 K_g)^{-1} \tilde{X}$, $B_\alpha^{-1} b_\alpha = \tilde{X}'(sI_n + \sigma^2 K_g)^{-1} y$, $\Sigma_0 = \text{diag}([v_{\alpha,0} \ \tau \lambda_1 \ \dots \ \tau \lambda_{m-1}]')$, $\tilde{X}$ is a n -by- m matrix stacking $\tilde{x}'_{L+1}, \dots, \tilde{x}'_T$ with $\tilde{x}'_t = [1 \ y_{t-1} \ \dots \ y_{t-L} \ x'_t]$ for $t = L+1, \dots, T$, I_n is a n -by- n identity matrix, $\tau = \tau^{(j-1)}$, $\lambda_i = \lambda_i^{(j-1)}$ for $i = 1, \dots, m-1$, $s = s^{(j)}$, $\sigma = \sigma^{(j-1)}$, and K_g is computed based on $\psi^{(j-1)}$.

(d) Use a Metropolis-Hastings step to draw $\sigma^{(j)}$ with the adaptive optimal scaling method of Garthwaite et al. (2016):

i. Draw a proposal σ^* from $N\left(\sigma^{(j-1)}, \left(v_\sigma^{(j-1)}\right)^2\right)$.

ii. Compute $p = \min\left(1, \frac{p(\sigma^*)}{p(\sigma^{(j-1)})}\right)$, where

$$p(\sigma) = \det(sI_n + \sigma^2 K_g)^{-\frac{1}{2}} \exp\left(-\frac{\sigma^2}{2\tau\lambda_m} - \frac{1}{2}(y - \tilde{X}\alpha)'(sI_n + \sigma^2 K_g)^{-1}(y - \tilde{X}\alpha)\right),$$

$s = s^{(j)}$, K_g is computed based on $\psi^{(j-1)}$, $\tau = \tau^{(j-1)}$, $\lambda_m = \lambda_m^{(j-1)}$, $\alpha = \alpha^{(j)}$.

iii. Draw u from a uniform distribution $U(0, 1)$. If $u \leq p$, set $\sigma^{(j)} = \sigma^*$ to accept the proposal σ^* . Otherwise, set $\sigma^{(j)} = \sigma^{(j-1)}$ to reject the proposal σ^* .

iv. Update $v_\sigma^{(j)} = v_\sigma^{(j-1)} \exp((p - p^*)/(p^*(1 - p^*)j))$, where $p^* = 0.44$ is the target acceptance probability of σ^* , $v_\sigma^{(0)}$ is a fixed pre-determined number (e.g. 0.01).

(e) Use a Metropolis-Hastings step to draw $\tilde{\psi}^{(j)}$ with the adaptive optimal scaling method of Garthwaite et al. (2016):

i. Draw a proposal $\tilde{\psi}^*$ from $N\left(\tilde{\psi}^{(j-1)}, \left(v_\psi^{(j-1)}\right)^2\right)$.

ii. Compute $p = \min \left(1, \frac{p(\tilde{\psi}^*)}{p(\tilde{\psi}^{(j-1)})} \right)$, where

$$p(\tilde{\psi}) = \det(s\mathbf{I}_n + \sigma^2 K_g)^{-\frac{1}{2}} \exp \left(-\frac{\tilde{\psi}^2}{2\lambda_\psi} - \frac{1}{2}(y - \tilde{X}\alpha)'(s\mathbf{I}_n + \sigma^2 K_g)^{-1}(y - \tilde{X}\alpha) \right),$$

$$s = s^{(j)}, \sigma = \sigma^{(j)}, \lambda_\psi = \lambda_\psi^{(j-1)}, \alpha = \alpha^{(j)}.$$

iii. Draw u from a uniform distribution $U(0, 1)$. If $u \leq p$, set $\tilde{\psi}^{(j)} = \tilde{\psi}^*$ to accept the proposal $\tilde{\psi}^*$. Otherwise, set $\tilde{\psi}^{(j)} = \tilde{\psi}^{(j-1)}$ to reject the proposal $\tilde{\psi}^*$.

iv. Update $v_\psi^{(j)} = v_\psi^{(j-1)} \exp((p - p^*)/(p^*(1 - p^*)j))$, where $p^* = 0.44$ is the target acceptance probability of $\tilde{\psi}^*$, $v_\psi^{(0)}$ is a fixed pre-determined number (e.g. 0.01).

(f) Draw $g^{(j)}$ from $N(b_g, B_g)$, where $B_g^{-1} = K_g^{-1} + \frac{\sigma^2}{s}\mathbf{I}_n$, $B_g^{-1}b_g = \frac{\sigma}{s}(y - \tilde{X}\alpha)$, K_g is computed based on $\tilde{\psi}^{(j)}$, $s = s^{(j)}$, $\sigma = \sigma^{(j)}$, $\alpha = \alpha^{(j)}$.

(g) Implement the ancillarity-sufficiency interweaving strategy (Yu and Meng (2011)) to boost the sampling efficiency:

i. Compute $f^* = \sigma^{(j)}g^{(j)}$.

ii. Keep the sign of $\sigma^{(j)}$ as Sign.

iii. Draw σ^{*2} from $GIG\left(\frac{1-n}{2}, \frac{1}{\tau\lambda_m}, f^{*'}K_g^{-1}f^*\right)$, where $GIG(\cdot, \cdot, \cdot)$ denotes the generalized inverse Gaussian distribution, $\tau = \tau^{(j-1)}$, $\lambda_m = \lambda_m^{(j-1)}$, K_g is computed based on $\tilde{\psi}^{(j)}$.

iv. Update $\sigma^{(j)} = \sqrt{\sigma^{*2}} \text{Sign}$.

v. Update $g^{(j)} = f^*/\sigma^{(j)}$.

(h) Draw $\tau^{(j)}$ from $IG((1+m)/2, 1/\tau_0 + \beta' \text{diag}(1/\lambda)\beta/2)$, where $\tau_0 = \tau_0^{(j-1)}$, $\beta = \beta^{(j)}$, $\lambda = \lambda^{(j-1)}$.

(i) Draw $\tau_0^{(j)}$ from $IG(1, n^2 + 1/\tau)$, where $\tau = \tau^{(j)}$.

(j) Draw $\lambda_i^{(j)}$ from $IG(1, 1/\lambda_{0,i} + (\beta' \mathbf{1}_i)^2/(2\tau))$, where $\lambda_{0,i} = \lambda_{0,i}^{(j-1)}$, $\beta = \beta^{(j)}$, $\mathbf{1}_i$ is a m -by-1 vector of zeros except for the i -th element being 1, $\tau = \tau^{(j)}$, $i = 1, \dots, m$. Assemble $\lambda^{(j)} = [\lambda_1^{(j)} \dots \lambda_m^{(j)}]'$.

- (k) Draw $\lambda_{0,i}^{(j)}$ from $IG(1, 1 + 1/\lambda_i)$, where $\lambda_i = \lambda_i^{(j)}$, $i = 1, \dots, m$. Assemble $\lambda_0^{(j)} = [\lambda_{0,1}^{(j)} \dots \lambda_{0,m}^{(j)}]'$.
- (l) Draw $\lambda_\psi^{(j)}$ from $IG(1, 1/\lambda_{\psi,0} + \tilde{\psi}^2/2)$, where $\lambda_{\psi,0} = \lambda_{\psi,0}^{(j-1)}$, $\tilde{\psi} = \tilde{\psi}^{(j)}$.
- (m) Draw $\lambda_{\psi,0}^{(j)}$ from $IG(1, 1 + 1/\lambda_\psi)$, where $\lambda_\psi = \lambda_\psi^{(j)}$.
- (n) Compute $f^{(j)} = \sigma^{(j)}g^{(j)}$, $\psi^{(j)} = \left(\tilde{\psi}^{(j)}\right)^2$.

4. Drop the initial M_0 sweeps as burn-ins and keep the next $M - M_0$ sweeps for analysis².

3 Prediction

For expositional convenience, define $a^{t,h} = [a_{t+1} \dots a_{t+h}]'$ as the outcome of a generic random variable a from time $t+1$ to $t+h$. Write $a^{0,h}$ as a^h for simplicity. Also, throughout the discussion the regressors x_t and z_t are treated as given to simplify notations.

The objective of prediction from the regularized kernel machine is to forecast $y^{t,h}$ based on observation of the history y^t . It is known in the statistics literature that the forecast minimizing the average squared forecast error is the conditional mean $E(y^{t,h}|y^t) \equiv \int y^{t,h} p(y^{t,h}|y^t) dy^{t,h}$ where $p(y^{t,h}|y^t)$ is the probability distribution of $y^{t,h}$ conditional on y^t . Using the law of total expectation leads to $E(y^{t,h}|y^t) = E(E(y^{t,h}|\theta, y^t)|y^t)$ which can be written as $\int E(y^{t,h}|\theta, y^t) p(\theta|y^t) d\theta$. Note that $p(\theta|y^t)$ is the posterior distribution of model parameters θ obtained in the estimation stage described in Section 2. The Monte Carlo approximation is:

$$\begin{aligned} E(y^{t,h}|y^t) &= \int E(y^{t,h}|\theta, y^t) p(\theta|y^t) d\theta \\ &\approx \frac{1}{M} \sum_{j=1}^M E(y^{t,h}|\theta^{(j)}, y^t) \end{aligned} \quad (3)$$

where $\{\theta^{(j)}\}_{j=1}^M$ are posterior draws of θ from $p(\theta|y^t)$.

²A rule of thumb in practice is to set M_0 to about 20% of M but it can vary depending on the assessment of the convergence of the draws.

To compute $E(y^{t,h}|\theta, y^t)$, re-write the kernel machine of Equation (1) in a vector form:

$$\begin{aligned} y_t &= \mathbf{1}'_1 \tilde{y}_t \\ \tilde{y}_t &= a + B\tilde{y}_{t-1} + Cx_t + d_t + \tilde{\epsilon}_t \end{aligned} \quad (4)$$

where $\mathbf{1}_1$ is a L -by-1 vector of zeros except for the first element being 1, $\tilde{y}_t = [y_t \ y_{t-1} \ \dots \ y_{t-L}]'$, a is a L -by-1 vector of zeros except for the first element being β_0 , B is a L -by- L matrix where the first row is $[\beta_1 \ \dots \ \beta_L]$, the $(i, i-1)$ element is one for $i = 2, \dots, L$, and all other elements are zeros, C is L -by- q matrix where q is the length of x_t , the first row is γ' , and all other elements are zeros, d_t is a L -by-1 vector of zeros except for the first element being $f(z_t)$, and finally $\tilde{\epsilon}_t$ is a L -by-1 vector of zeros except for the first element being ϵ_t .

From Equation (4), it is straightforward to derive:

$$\begin{aligned} E(y_{t+i}|\theta, y^t) &= \mathbf{1}'_1 E(\tilde{y}_{t+i}|\theta, y^t) \\ E(\tilde{y}_{t+i}|\theta, y^t) &= (I_L + B + \dots + B^{i-1})a + B^i \tilde{y}_t \\ &\quad + B^{i-1}Cx_{t+1} + B^{i-2}Cx_{t+2} + \dots + Cx_{t+i} \\ &\quad + B^{i-1}E(d_{t+1}|\theta, y^t) + B^{i-2}E(d_{t+2}|\theta, y^t) \\ &\quad + \dots + E(d_{t+i}|\theta, y^t) \end{aligned} \quad (5)$$

for $i = 1, \dots, h$. To compute the component $E(d_{t+i}|\theta, y^t)$, one can use the following equations:

$$\begin{aligned} E(f^{t,h}|\theta, y^t) &= A'\Sigma^{-1}E(f^{L,t-L}|\theta, y^t) \\ E(f^{L,t-L}|\theta, y^t) &= (s\Sigma^{-1} + I_{t-L})^{-1} \left(y^{L,t-L} - \tilde{X}^{L,t-L}\alpha \right) \end{aligned} \quad (6)$$

where A is the row-1-to-row- $(t-L)$ and column- $(t-L+1)$ -to-column- $(t-L+h)$ submatrix of the $(t-L+h)$ -by- $(t-L+h)$ matrix $\tilde{\Sigma}$ whose (i, j) element is $\sigma^2 \exp(-\psi(z_{L+i} - z_{L+j}))(z_{L+i} - z_{L+j})$ for $i, j = 1, \dots, t-L+h$, Σ is the row-1-to-row- $(t-L)$ and column-1-to-column- $(t-L)$ submatrix of $\tilde{\Sigma}$, $\tilde{X}^{L,t-L}$ is a $(t-L)$ -by- m matrix stacking $\tilde{x}'_{L+1}, \dots, \tilde{x}'_t$ with $\tilde{x}'_j = [1 \ y_{j-1} \ \dots \ y_{j-L} \ x'_j]$ for $j = L+1, \dots, t$.

In cases where the model target y_t is transformed from some economic target Y_t by $y_t = Y_t - Y_{t-1}$ and the ultimate interest is to predict Y_t^3 , one could use Equation (7) to

³For example, Y_t could be the revenue or the log asset size of a business unit.

compute $E(Y^{t,h}|\theta, y^t)$ and then apply Equation (3) to produce the forecast $E(Y^{t,h}|\theta, y^t)$.

$$E(Y_{t+i}|\theta, y^t) = Y_t + E(y_{t+1}|\theta, y^t) + \dots + E(y_{t+i}|\theta, y^t), \quad i = 1, \dots, h \quad (7)$$

In summary, the prediction steps are as follows:

1. Collect posterior draws $\{\theta^{(j)}\}_{j=1}^M$ from the estimation stage of Section 2.
2. Design a for loop with M sweeps.
3. Within sweep $j \in \{1, 2, \dots, M\}$, do the following:
 - (a) Use Equation (6) to compute $E(f^{t,h}|\theta^{(j)}, y^t)$.
 - (b) Use Equation (5) to compute $E(y^{t,h}|\theta^{(j)}, y^t)$.
 - (c) If the target is to forecast Y_t where $y_t = Y_t - Y_{t-1}$, use Equation (7) to compute $E(Y^{t,h}|\theta^{(j)}, y^t)$.
4. Use Equation (3) to compute $E(y^{t,h}|y^t)$ or $E(Y^{t,h}|\theta, y^t)$ as the final forecast.

4 Simulation Study

A small scale simulation study is conducted to check the performance of the regularized kernel machine. Two data generating processes (DGP) are considered that simulate from Equation (1) based on two choices of σ :

1. $\sigma = 0$ (i.e. the DGP is linear since $f(z_t) = 0$).
2. $\sigma = 1$ (i.e. the DGP is nonlinear).

The other parameters of Equation (1) are the same in the two DGPs with $T = 100$, $L = 3$, $\beta_0 = 0.1$, $\beta_1 = 0$, $\beta_2 = 0.9$, $\beta_3 = 0$, $\psi = 1$, $s = 0.1$, $\gamma = [1 \ 0]'$, $x_t \sim N(0, I_2)$ and $z_t = x_t$.

A total of 5,000 posterior draws are kept for analysis after 2,000 burn-ins. The posterior draws mix reasonably well as shown in Figure 1 for $\log(\sigma^2)$ in the two DGPs⁴. Figure 2

⁴The transformation $\log(\sigma^2)$ is used to assess mixing quality instead of σ since the sign of σ is unidentified in the kernel machine.

displays the posterior distributions of the key parameter σ in the two DGPs. For DGP 1, the posterior distribution of σ shows the correct uni-modal shape around zero, while the posterior distribution of σ is bimodal around the correct values for DGP 2. Table 1 shows that the posterior mean of the other model parameters in the two DGPs are close to their true values⁵.

The regularized kernel machine is compared against the OLS regression model in a out-of-sample predictive exercise. A total of 20 simulations are conducted. In each simulation, a sample of 100 data points are simulated from the two DGPs according to the description in the first paragraph of this section. 12 data points at the end of the sample are set aside as the prediction sample. Both the regularized kernel machine and the OLS regression model are estimated based on the remainder of the simulated data points. For the OLS regression model, the backward elimination procedure is used to select variables where the p-value threshold is 5%. The mean squared error (MSE) of the forecasts for the prediction sample from the two models are calculated for comparison.

Table 2 compares that average predictive MSE of the two models over the 20 simulations for both DGP 1 and DGP 2. Intuitively, for DGP 1, the OLS regression model is the correct one, while the nonlinear component $f(z_t)$ in the regularized kernel machine is redundant and could potentially lead to over-fitting. However, if the regularization through the shrinkage priors placed on its model parameters is effective, one should expect the regularized kernel machine to perform similarly as the OLS regression model. On the other hand, DGP 2 is nonlinear. Therefore, one should expect the regularized kernel machine to outperform the OLS regression model for DGP 2 as it allows for the possibility of nonlinearity. The results in Table 2 confirm these intuitions. The average predictive MSE of the regularized kernel machine is close to that of the OLS regression model for DGP 1 while being much smaller in DGP 2. Figure 3 plots the predictive MSEs of the two models in the 20 simulations and further confirms the findings in Table 2.

⁵Given the strong serial correlation due to $\beta_2 = 0.9$, there is considerable uncertainty about the intercept β_0 reflected by its wide credible set.

Table 1: Parameter Estimate: Regularized Kernel Machine

		DGP 1		DGP 2	
	True Value	Estimate	True Value	Estimate	
β_0	0.1	0.05 (-0.04, 0.15)	0.1	0.67 (-0.21, 1.61)	
β_1	0	0.01 (-0.02, 0.05)	0	0.02 (-0.02, 0.07)	
β_2	0.9	0.89 (0.87, 0.92)	0.9	0.89 (0.85, 0.93)	
β_3	0	-0.01 (-0.05, 0.01)	0	-0.03 (-0.09, 0.01)	
γ_1	0	0.99 (0.94, 1.04)	1	0.70 (-0.00, 1.19)	
γ_2	0	0.01 (-0.02, 0.06)	0	0.16 (-0.10, 0.57)	
ψ			1	0.48 (0.18, 1.10)	
s	0.1	0.08 (0.06, 0.11)	0.1	0.15 (0.10, 0.20)	

Note: The "Estimate" column shows the posterior mean and the 90% credible set in brackets.

Table 2: Comparing Average Predictive MSE

	OLS Regression	Regularized Kernel Machine
DGP 1	0.29	0.30
DGP 2	2.34	0.64

Note: The table shows the average predictive MSE of the OLS regression model and the regularized kernel machine over 20 simulations from DGP 1 and DGP 2.

Figure 1: Posterior Draws of $\log(\sigma^2)$

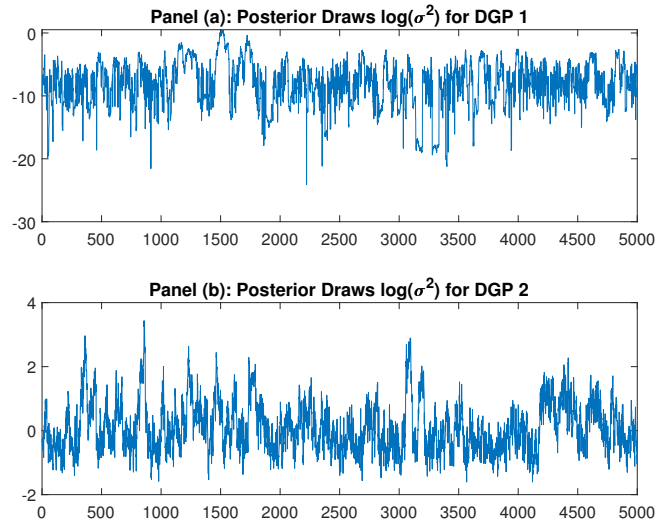


Figure 2: Posterior Distributions of σ

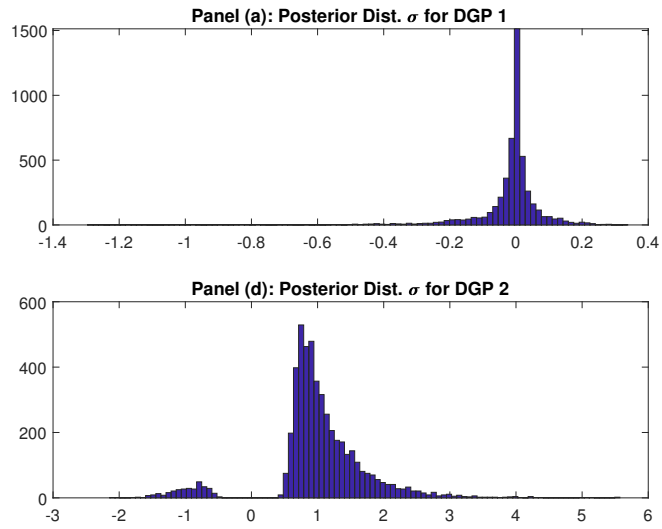
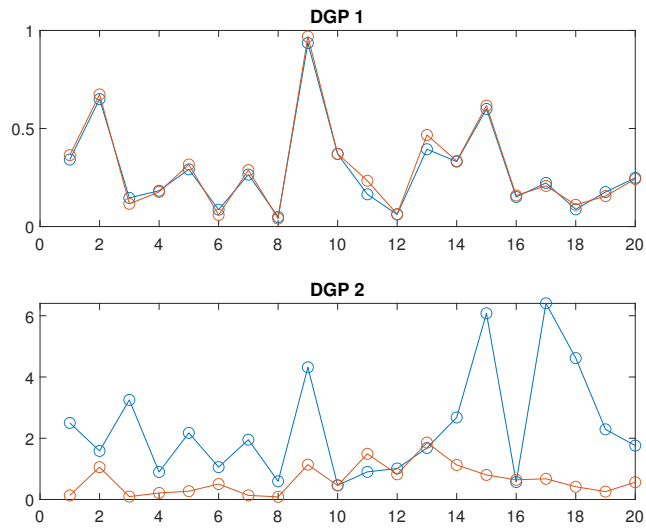


Figure 3: Comparing Predictive MSEs



Note: The blue line shows the predictive MSEs of the OLS regression model over 20 simulated data samples, while the red line shows those of the regularized kernel machine. A smaller MSE indicates better predictive performance.

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